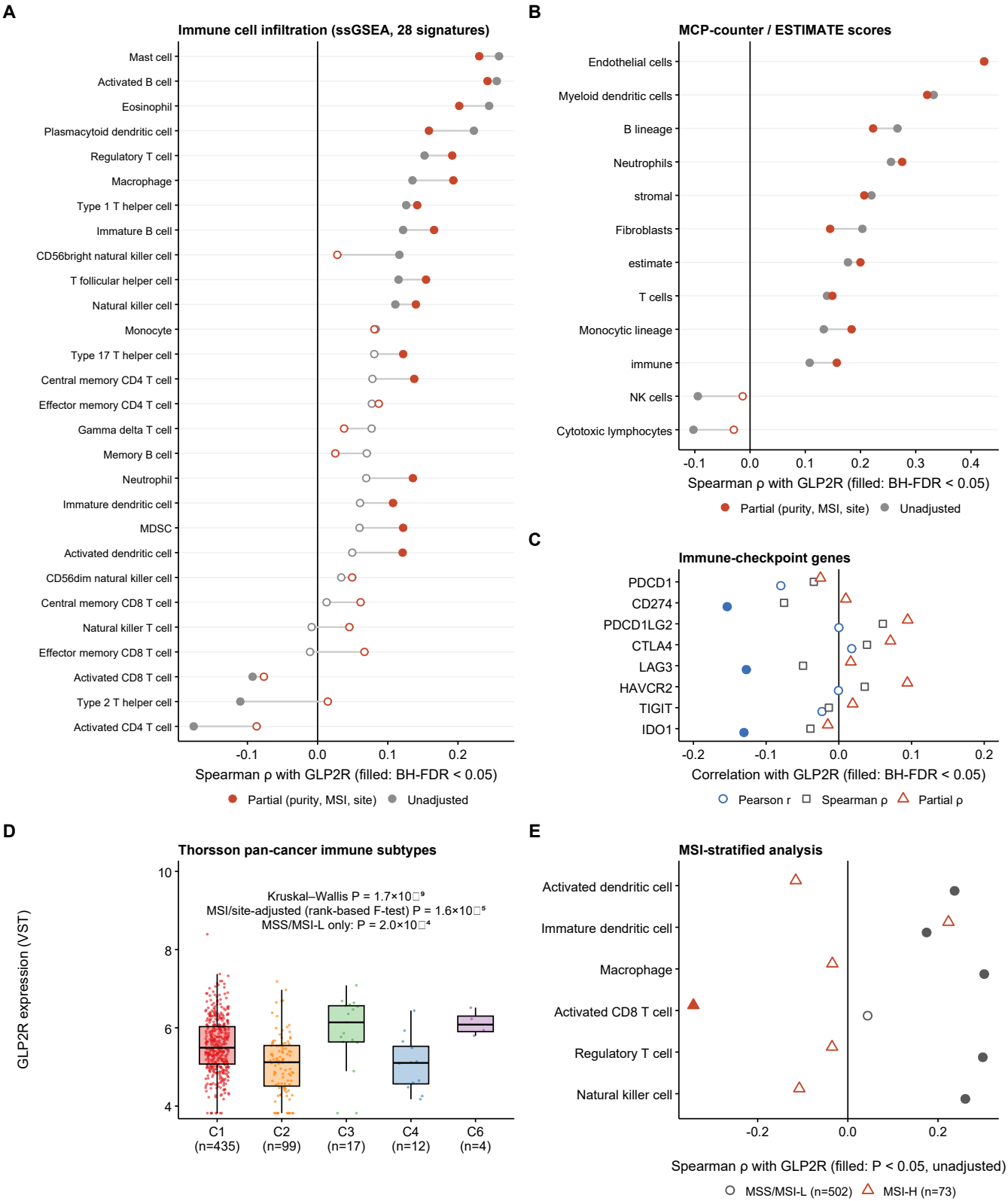

A
Immune cell infiltration (ssGSEA, 28 signatures)
Mast cell
Activated B cell
Eosinophil
Plasmacytoid dendritic cell
Regulatory T cell
Macrophage
Type 1 T helper cell
Immature B cell
CD56bright natural killer cell
T follicular helper cell
Natural killer cell
Monocyte
Type 17 T helper cell
Central memory CD4 T cell
Effector memory CD4 T cell
Gamma delta T cell
Memory B cell
Neutrophil
Immature dendritic cell
MDSC
Activated dendritic cell
CD56dim natural killer cell
Central memory CD8 T cell
Natural killer T cell
Effector memory CD8 T cell
Activated CD8 T cell
Type 2 T helper cell
Activated CD4 T cell
-0.1
0.0
0.1
0.2
Spearman ρ with GLP2R (filled: BH-FDR < 0.05)
Partial (purity, MSI, site)
Unadjusted
B
MCP-counter / ESTIMATE scores
Endothelial cells
Myeloid dendritic cells
B lineage
Neutrophils
stromal
Fibroblasts
estimate
T cells
Monocytic lineage
immune
NK cells
Cytotoxic lymphocytes
-0.1
0.0
0.1
0.2
0.3
0.4
Spearman ρ with GLP2R (filled: BH-FDR < 0.05)
Partial (purity, MSI, site)
Unadjusted
C
Immune-checkpoint genes
PDCD1
CD274
PDCD1LG2
CTLA4
LAG3
HAVCR2
TIGIT
IDO1
-0.2
-0.1
0.0
0.1
0.2
Correlation with GLP2R (filled: BH-FDR < 0.05)
Pearson r
Spearman ρ
Partial ρ
D
Thorsson pan-cancer immune subtypes
GLP2R expression (VST)
10
8
6
4
Kruskal–Wallis P = 1.7×10□⁹
MSI/site-adjusted (rank-based F-test) P = 1.6×10□⁵
MSS/MSI-L only: P = 2.0×10□⁴
C1
(n=435)
C2
(n=99)
C3
(n=17)
C4
(n=12)
C6
(n=4)
E
MSI-stratified analysis
Activated dendritic cell
Immature dendritic cell
Macrophage
Activated CD8 T cell
Regulatory T cell
Natural killer cell
-0.2
0.0
0.2
Spearman ρ with GLP2R (filled: P < 0.05, unadjusted)
MSS/MSI-L (n=502)
MSI-H (n=73)